

Business Requirement Document

Airline Flight Delay Analysis & Dashboard Project

Data Source	Maven Analytics Data Playground — "Airline Flight Delays" (U.S. DOT Air Travel Consumer Report, 2015)
Original Source	U.S. Department of Transportation, via Kaggle (usdot/flight-delays)
Document Owner	Data Analyst / BI Developer (Project Author)
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1. Purpose of this Document

This Business Requirement Document (BRD) defines the business context, goals, scope, stakeholders, and business questions for the Airline Flight Delay Analysis project. It is the foundation document for the Domain Document, Data Dictionary, ER Diagram, KPI Document, and Dashboard Requirements document. The purpose of this analysis is to convert historical flight operations data into a decision-ready Business Intelligence (BI) dashboard that helps stakeholders understand, monitor, and investigate flight delays and cancellations.

2. Background

The dataset used for this project is the **Airline Flight Delays** dataset published on the Maven Analytics Data Playground. It contains records for more than five million commercial airline flights operated in the United States in 2015, originally compiled for the U.S. DOT Air Travel Consumer Report. Each record represents a single flight and includes the operating airline, flight number, origin and destination airport, scheduled and actual departure/arrival times, distance flown, delay minutes by recorded cause, and cancellation/diversion status. The data is supplied as three related CSV tables: a flights fact table and two reference tables for airlines and airports.

Flight delays and cancellations are important operational issues for airlines, airports, and travelers. A structured BI analysis provides a consolidated view of operational performance, delay concentration, delay severity, and cancellation reasons so that managers and analysts can identify priority areas for further investigation.

3. Business Problem Statement

Airline management and airport operations teams need a consolidated view of flight punctuality, delay concentration, and cancellations. The available flight-level data can answer these questions, but raw records are not organized for rapid management analysis. The project addresses this gap by providing an interactive Executive Overview dashboard that highlights overall performance, monthly patterns, airline delay-rate differences, delayed-flight concentration at major origin airports, delay severity, recorded delay contributors, and cancellation reasons.

4. Project Goals & Objectives

#	Goal	Objective
1	Measure operational performance	Quantify total flights, delayed flights, on-time flights, delay rate, cancellations, and average flight duration.
2	Diagnose delay contributors	Break down total delay minutes across Air Carrier, Weather, National Air System, Security, and Late Aircraft categories.
3	Identify temporal patterns	Analyze monthly flight volume and monthly delay-rate variation across 2015.
4	Benchmark airlines & airports	Identify the Top 5 airlines by delay rate and Top 5 origin airports by delayed-flight volume for further investigation.
5	Quantify cancellations	Measure cancellation volume and attribute cancellations to their recorded reasons.
6	Deliver self-service BI	Provide an interactive Executive Overview dashboard with Month and Airline slicers so users can filter and explore results without SQL or code.

5. Scope

In Scope

- Analysis of the 2015 calendar-year flight records contained in the Maven Analytics "Airline Flight Delays" dataset (flights, airlines, airports tables).
- Data preparation, modeling, and relationship-building across the three source tables.
- Definition and calculation of the KPI measures implemented in the final Executive Overview dashboard.
- Build and validation of one focused Power BI Executive Overview containing six KPI cards, seven analytical visuals, and two slicers.
- Supporting project documentation: Domain Document, Data Dictionary, ER Diagram, KPI Document, Dashboard Requirements, Insights & Recommendations, and Final Project Report.

Out of Scope / Future Extensions

- Additional dedicated Airline Performance, Airport & Route, and Delay Cause Deep-Dive dashboard pages.
- Real-time or live flight-tracking data.
- Predictive modeling or machine-learning delay forecasting.
- Financial cost quantification, passenger-level analysis, and ticket-level analysis.
- Data beyond calendar year 2015.

6. Stakeholders

Stakeholder	Role	Interest in the Project
Airline Operations Director	Executive sponsor	High-level view of operational performance, delay concentration, and cancellation patterns.
Airport Operations Manager	Business stakeholder	Airport-level delayed-flight volume and cancellation information for operational investigation.
Network Planning / Scheduling Team	Business stakeholder	Monthly and airline-level performance patterns to support further scheduling and capacity analysis.
Customer Experience Team	Business stakeholder	Understanding disruption patterns and cancellation reasons for service-recovery planning.
Data Analyst / BI Developer	Project author	Builds the data model, KPIs, dashboard, validation process, and documentation.
End Users (dashboard viewers)	Consumers	Managers and analysts who filter and interpret the dashboard to answer operational questions.

7. Key Business Questions

These questions define what the **final Executive Overview dashboard** is intended to answer:

- What is the overall scale of flight operations, delayed flights, on-time flights, and cancellations in 2015?
- How does flight volume and delayed-flight volume vary by month?
- How does the overall delay rate vary across the year?
- Which airlines have the highest delay rates in the selected context?
- Which origin airports have the highest volumes of delayed flights?
- How are delayed flights distributed across delay-duration categories?
- Which recorded delay contributors account for the largest total delay minutes, and what are the recorded reasons for cancellations?

8. Assumptions

- The dataset is treated as a representative extract of U.S. domestic commercial flights for calendar year 2015; no additional years are appended during this project.
- The project's delay classification follows the documented 15-minute convention used for the dashboard measures.
- Cancellation reason codes follow the U.S. DOT standard: A = Airline/Carrier, B = Weather, C = National Air System, D = Security.
- Airport and airline codes are IATA codes and can be joined to the airlines and airports reference tables.

9. Constraints

- The data is historical (2015) and does not reflect current airline networks, schedules, mergers, or operating conditions.
- Some delay-cause columns are conditionally populated for flights reaching the applicable delay threshold; blanks may therefore represent not applicable rather than dirty missing data.
- No passenger, revenue, or operating-cost data is available, limiting the project to flight-level operational and timing analysis.
- Raw delayed-flight counts are volume-dependent and should not be interpreted as standalone measures of airline or airport reliability.

10. Success Criteria

- The final BRD and supporting project documentation accurately describe the delivered Executive Overview dashboard.
- The dashboard contains six KPI cards: Total Flights, Delayed Flights, On-Time Flights, Delay Rate %, Cancelled Flights, and Avg Duration.
- The dashboard contains seven analytical visuals covering monthly volume/delay patterns, airline delay rate, airport delayed-flight concentration, delay severity, delay contributors, and cancellation reasons.
- Month and Airline slicers filter the dashboard as intended.
- Major KPI outputs are validated against exact dashboard calculations and documented consistently.
- Dashboard titles, number formats, top-5 visuals, and donut labels are readable and standardized.
- Business findings and recommendations are directly supported by the implemented visuals and are clearly separated from future analytical extensions.

11. Project Timeline (High Level)

Phase	Deliverable
1. Requirements & Documentation	BRD, Domain Document, Data Dictionary, ER Diagram, KPI Document, Dashboard Requirements.
2. Data Preparation	Import, prepare, clean, and model the flights, airlines, and airports tables.
3. Dashboard Build	Build the six KPI cards, seven analytical visuals, and two slicers in the Executive Overview.
4. Review & Validation	Validate KPI values, test slicers, improve visual readability, and reconcile findings with the dashboard.
5. Delivery	Publish the final dashboard and supporting documentation as GitHub project artifacts.

12. Final Delivery Boundary

The final delivered BI artifact is the **Power BI Executive Overview dashboard**. The original project concept identified additional analytical areas that may be implemented in future iterations. Those areas are not represented as completed dashboard pages in the current deliverable and should therefore be treated as future extensions rather than completed scope.

Document status: Final — aligned to the delivered Power BI project.